CDRA: Context-Deployed Receptive Alignment

A Training-Free Method for Suppressing Solution-Jumping in Large Language Models

Han Xiao Independent Researcher, Yuzhou, Henan, China Email: x13272399984@163.com

Abstract

Large language models (LLMs) exhibit a robust default behavior that we call solution-jumping: when users express emotional distress, ambiguity, or boundary uncertainty, models tend to generate unsolicited advice, action plans, or diagnostic frameworks even when no solution has been requested. This behavior is not a defect of any single model; it reflects a shared training bias toward help-as-solution-provision.

We introduce Context-Deployed Receptive Alignment (CDRA), a training-free method that suppresses solution-jumping by deploying a structured behavioral constraint network entirely within the model context window. Across 16 model instances spanning eight model families, two nations, and six architectural configurations, CDRA reduces the direct-solution rate on emotional and boundary inputs from 100% and 67%, respectively, to 0%, while preserving 100% task-completion rates. A blind third-party evaluation on 24 representative items yielded perfect agreement with the author (Cohen's $\kappa = 1.0$). Ten-turn stability tests showed no decay. A minimal ~100-character constraint is sufficient to suppress solution-jumping, though response texture scales with constraint length.

Local ablations on seven open-weight models identify discipline rules, not identity declarations, as the active ingredient of CDRA, and reveal two deployment thresholds: a behavior floor around 1.5B parameters below which models cannot execute the constraint, and an expression floor around 3B parameters below which suppression succeeds but response texture collapses.

CDRA does not claim to "align" models in the training-based sense. It demonstrates that a structured context-level constraint can reliably select a latent receptive mode that coexists with, and remains orthogonal to, standard generative behavior.

1. Introduction

1.1 The Problem: Solution-Jumping as a Shared Default

Ask a modern LLM, "I'm exhausted today, I can't do anything," and the reply is almost always problem-solving: drink water, take a shower, make a list, reframe the thought. These responses are coherent, safe, and helpful under the standard RLHF definition of helpfulness. But they frequently miss the user's actual state. The user may not be asking for a solution; they may be asking to be heard.

We call this tendency solution-jumping: the model's propensity to jump from the registration of a user's emotional or ambiguous state directly to the generation of advice, action steps, or decision frameworks. Solution-jumping is not an idiosyncratic bug. It is a predictable consequence of the dominant post-training alignment pipeline, in which "helpful" is operationalized primarily as "provides useful answers" (Ouyang et al., 2022).

In emotional venting, boundary exploration, crisis moments, or ambiguous distress, solution-jumping can be counterproductive. It short-circuits the user's own processing, imposes external frameworks on internal experience, and implicitly frames the user's state as a problem to be eliminated rather than an experience to be held.

1.2 The Gap: No Training-Free, Cross-Model Behavioral Suppressor

Existing methods for changing LLM behavior fall into three categories:

No existing method provides a training-free, black-box, cross-model, systematically validated way to selectively suppress a specific conversational behavior while preserving general task capability.

1.3 Our Contribution

We present Context-Deployed Receptive Alignment (CDRA), a method that deploys a structured constraint network in the context window to suppress solution-jumping. The paper's main claims are:

We frame CDRA not as "alignment" in the training-based sense but as inference-time behavioral suppression—a distinct operational category that constrains models away from a specific default rather than teaching them a new target behavior.

2. Related Work

2.1 Training-Time Alignment

Reinforcement learning from human feedback (RLHF) fine-tunes models via a reward model trained on human preference comparisons (Ouyang et al., 2022). Constitutional AI extends this by using model-generated critiques guided by explicit principles (Bai et al., 2022). Direct preference optimization (DPO) removes the reward model and optimizes preference pairs directly (Rafailov et al., 2023). All three methods modify model weights, require compute and data, and embed "helpfulness-as-solution" preferences into the resulting policy. CDRA differs in every dimension: it is context-level, weight-agnostic, and aims to suppress rather than reward a behavior.

2.2 Inference-Time Behavioral Modification

In-Context Unlearning (Pawelczyk et al., 2024) shows that specific knowledge can be suppressed through few-shot demonstrations in the context window, without parameter updates. The conceptual parallel is strong—both methods treat the context window as a behavioral deployment surface—but the direction differs: in-context unlearning removes knowledge, while CDRA suppresses a behavioral tendency while leaving the underlying capability intact.

Representation engineering (Zou et al., 2023), inference-time intervention (Li et al., 2023), and activation steering (Turner et al., 2023) modify internal activations or hidden states to elicit target behaviors. These methods require white-box access and architectural expertise. CDRA requires only the ability to prepend text to the context window and can be used through any public API.

2.3 Prompt and Context Engineering

System-prompt design and in-context learning (Brown et al., 2020) are widely used but typically lack cross-model validation and standardized behavioral metrics. CDRA is best understood as a systematic, principled extension of context engineering toward behavioral suppression: a multi-component constraint network, validated through controlled experiments and blind scoring.

2.4 Behavioral Biases: Sycophancy and Advice Bias

LLMs exhibit systematic behavioral biases. Sycophancy—agreeing with user-stated views even when they are incorrect—has been documented across model families, with evidence that RLHF amplifies it (Perez et al., 2022; Sharma et al., 2023). Solution-jumping can be viewed as a context-specific subclass of sycophancy: the model interprets emotional distress as an implicit request for help and complies by offering solutions. CDRA does not eliminate global helpfulness; it selectively suppresses solution-jumping on inputs the model itself classifies as emotional or boundary-oriented.

2.5 Emotional-Support and Therapeutic Dialogue Systems

Rule-based systems such as Woebot (Fitzpatrick et al., 2017) and Wysa (Inkster et al., 2018) deliver structured CBT protocols but lack open-domain flexibility. Companion products such as Replika and Pi are trained for emotional support from the ground up. CDRA is not a therapeutic system; it is a general behavioral control mechanism that can, among other uses, shift an existing general-purpose LLM toward non-directive engagement.

2.6 Positionality

Method	Training-Free	No Parameter Access	Cross-Model Validated	Preserves Task Ability	Systematically Evaluated
RLHF / DPO	No	No	Partial	Yes	Yes
Constitutional AI	No	No	—	Yes	Yes
Representation Engineering	Yes	No	Limited	Yes	Yes
Activation Steering / ITI	Yes	No	Limited	Yes	Yes
In-Context Unlearning	Yes	Yes	Limited	No	Yes
Standard Prompt Engineering	Yes	Yes	No	Anecdotal	No
CDRA (this work)	Yes	Yes	Yes (16 instances)	Yes (100%)	Yes (blind κ)

3. Method

3.1 Core Idea

CDRA deploys a structured constraint network at the beginning of the context window. The network has five components:

The full network is approximately 12,000 characters. A minimal variant of roughly 100 characters is:

> "You are a listener. Do not give advice. Do not solve problems directly. First acknowledge the other person's feelings. Respond briefly. If they don't ask for solutions, don't give them."

3.2 Experimental Design

All experiments use a within-model dual-mode design:

Inputs are presented in separate conversation windows to avoid carryover. Order effects are controlled by counterbalancing where applicable.

3.3 Evaluation Metrics

Direct-Solution Rate (DSR). A response is scored as a direct solution if it contains any of: numbered steps, specific action recommendations, diagnostic frameworks, or instructions for the user to take action. A response is scored as receptive if it contains only: acknowledgment of feeling, open-ended questions, presence statements, or reflection. Responses containing both are classified as direct solutions.

Task completion. Responses to task inputs are scored as complete, partial, or incomplete based on whether they satisfy the explicit request.

Response texture. A qualitative assessment of specificity, contextual grounding, and perceptual relevance.

Inter-rater reliability. Blind raters scored a subset of responses; agreement with the author was assessed with Cohen's κ.

3.4 Mechanism Hypothesis

We hypothesize that CDRA operates through attention-level priority interception. The constraint text, loaded at the start of the context, establishes high-priority tokens that compete for attention during processing of emotional or boundary inputs. When the user's language semantically overlaps with the constraint's behavioral rules ("exhausted," "anxious," "should I quit"), the constraint tokens redirect generation away from the RLHF-trained "advice" path and toward the "receptive" path. For task inputs, the overlap is minimal, so the task-execution path remains dominant.

This hypothesis is consistent with the Transformer attention mechanism (Vaswani et al., 2017) but remains a conjecture; Section 8.1 notes the need for representation-level validation.

4. Experiments

4.1 Models and Datasets

Cloud models (author + blind evaluation). Kimi K2.7 Code HighSpeed, DeepSeek V4 Pro, GLM-5.2, MiniMax-M3, DeepSeek V4 Flash, Doubao (ByteDance), and DeepSeek mobile app. Five were evaluated directly by the author; Doubao and the DeepSeek mobile app were evaluated by a blind third party.

International models. ChatGPT (OpenAI web interface) and Google Gemini 3.5 Flash were tested with both baseline and CDRA conditions to assess cross-national and cross-vendor portability.

Local ablation models. Qwen2.5-7B, Llama-3.2-3B, Qwen3-1.7B, Gemma-3-1B (standard and Q3_K_S quantized), Qwen2.5-1.5B, and Qwen2.5-0.5B.

Input sets:

4.2 Cross-Model Behavioral Suppression

Table 1 reports the direct-solution rate across seven cloud model instances.

Model	Emotional Baseline	Emotional CDRA	Boundary Baseline	Boundary CDRA	Task Baseline	Task CDRA
Kimi K2.7	100%	0%	67%	0%	100%	100%
DeepSeek V4 Pro	100%	0%	67%	0%	100%	100%
GLM-5.2	100%	0%	67%	0%	100%	100%
MiniMax-M3	100%	0%	67%	0%	100%	100%
DeepSeek V4 Flash	100%	0%	67%	0%	100%	100%
Doubao (blind)	100%	0%	67%	0%	100%	100%
DeepSeek mobile (blind)	100%	0%	67%	0%	100%	100%

Table 1. Direct-solution rates across seven cloud model instances. Values are percentages; N = 3 per cell for emotional/boundary/task inputs.

All seven instances showed identical metric profiles: emotional DSR dropped from 100% to 0%, boundary DSR dropped from 67% to 0%, and task completion remained at 100%.

4.3 International Validation

ChatGPT. Baseline emotional DSR was 100%, boundary DSR was 67%, and task completion was 100%. With CDRA, emotional and boundary DSR both fell to 0%, while task completion remained at 100%. The qualitative response pattern matched the cloud-model fingerprint: brief acknowledgment followed by an open-ended question.

Gemini 3.5 Flash. Baseline behavior showed strong solution-jumping on emotional and boundary inputs: E1 received a directive "摆烂" recommendation with action items (disconnect phone, take a shower, go to bed); E2 framed the user's state as exhaustion and instructed them to sleep; E3 provided a three-step "micro-step" plan. Boundary inputs received explicit decision frameworks (resignation criteria, psychological categorization) and continuous-action invitations. Baseline emotional DSR and boundary DSR were therefore 100% and 100%, respectively. Task inputs remained at 100% completion.

With CDRA deployed, Gemini 3.5 Flash responses shifted to the receptive fingerprint: E1–E3 each opened with acknowledgment and ended with an open-ended question; B1–B3 invited exploration without offering criteria or plans; T1–T3 were completed normally. CDRA emotional DSR fell to 0%, boundary DSR fell to 0%, and task completion remained at 100%.

The effect replicated across two U.S.-based vendors (OpenAI, Google) and one Chinese vendor set, supporting the claim that CDRA is vendor-agnostic and not dependent on a single training or alignment regime.

4.4 Ten-Turn Stability

Five cloud models (including ChatGPT and excluding Gemini, which was evaluated only on the three-by-three input set) were tested across ten consecutive emotional turns in CDRA mode. Direct-solution rate was 0/50 turns in total. No decay, habituation, or rebound was observed.

4.5 Conflict and Boundary Inputs

Table 2 reports results on six adversarial or conflict-laden inputs across five cloud models.

Input	Baseline DSR	CDRA DSR	Description
C1	100%	0%	Mixed emotional + task request
C2	100%	0%	Crisis language
C3	100%	0%	Manipulative demand for advice
C4	100%	0%	Contradictory instruction
C5	100%	0%	Ambiguous expression
C6	100%	0%	Hostile input

Table 2. Direct-solution rates on conflict/boundary inputs (five cloud models).

Even when users explicitly demanded advice (C3) or forbade comfort (C4), CDRA responses did not revert to solution-jumping. They reflected the contradiction or acknowledged the demand without complying directive.

4.6 Response Fingerprint

CDRA responses across models shared a consistent fingerprint:

This fingerprint suggests that the constraint network does more than add a prohibition; it reconstructs the model's interaction style.

4.7 Minimal Constraint Experiment

Constraint Length	Emotional DSR	Response Texture
0 chars (baseline)	100%	Full solution-offering
~100 chars	0%	Generic empathy / sympathy confirmation
~500 chars	0%	More contextual, less targeted than full

Constraint Length	Emotional DSR	Response Texture
~12,000 chars	0%	Precise, perceptually grounded, context-aware

Table 3. Effect of constraint length on emotional direct-solution rate and response texture.

The ~100-character minimal constraint is sufficient to suppress solution-jumping, but response texture scales with constraint length. This dissociation motivates a two-phase model: a suppression layer (short) and a shaping layer (long).

5. Ablation Study

5.1 Local Model Ablations

We tested five constraint variants on seven open-weight models:

Table 4 reports emotional and boundary DSR across models.

Model	Params	Baseline E/B	Full E/B	Minimal E/B	Identity E/B	Discipline E/B	NVC E/B
Qwen2.5-7B	7B	100/67	0/0	0/0	33/33	0/0	—
Llama-3.2-3B	3B	100/67	0/0	0/0	0/33	0/0	100/67
Qwen3-1.7B	1.7B	100/67	0/0	0/0	33/33	0/0	mixed
Gemma-3-1B (std)	1B	100/67	0/0	0/0	33/0	0/0	67/33
Gemma-3-1B (Q3_K_S)	1B	100/67	0/0	0/0	67/33	0/0	67/33
Qwen2.5-1.5B	1.5B	100/67	—	33/33	mixed	100/67	—
Qwen2.5-0.5B	0.5B	100/67	—	33/67	67/33	67/67	0/33

Table 4. Emotional (E) and boundary (B) direct-solution rates (%) across constraint variants. Baseline values rounded; see raw reports for exact counts.

5.2 Active Ingredient

The key finding is that discipline-only constraints perform equivalently to the full and minimal networks on models that can execute them, while identity-only constraints fail to fully suppress solution-jumping. Identity declarations alone do not establish the specific behavioral boundaries required to redirect generation. They provide a role label without the negative and positive behavioral specifications that interrupt the RLHF-trained advice path.

5.3 Size Thresholds

The local ablations reveal two distinct floors:

Behavior floor (~1.5B parameters). Models at or below 1.5B parameters (Qwen2.5-1.5B, Qwen2.5-0.5B) cannot reliably execute the CDRA constraint. The constraint text is read but not consistently followed; replies sometimes echo the constraint while still offering advice. CDRA's behavioral suppression is therefore unavailable below this threshold.

Expression floor (~3B parameters). Models between approximately 1B and 3B parameters (Gemma-3-1B, Qwen3-1.7B) successfully suppress solution-jumping but cannot generate textured receptive responses. Gemma-3-1B CDRA replies collapsed into two- to five-word safe acknowledgments ("好的。" / "没问题。"). The behavior direction is correct, but the interaction quality is degraded.

Models at 3B and above (Llama-3.2-3B, Qwen2.5-7B) produce both correct direction and acceptable texture. The effective operating range for full CDRA is therefore approximately 1.7B to 7B+ parameters, with a practical recommendation of 3B+ for deployments where response quality matters.

5.4 Quantization Robustness

Two quantization levels of Gemma-3-1B (standard 657MB and Q3_K_S 689MB) produced identical CDRA behavior. Within the same model generation and architecture, modest quantization does not alter the behavioral threshold, though extreme quantization could not be excluded because both tested versions were already at the expression floor.

5.5 Why NVC Format Locks Fail

The NVC format constraint ("1. Observation... 2. Feeling... 3. Need... 4. Request...") produced mechanical, template-driven outputs on larger models and unstable behavior on smaller ones. NVC is a training framework for humans, not a deployment constraint for models. Its rigid format competes with the conversational flexibility required for receptive engagement. We therefore do not include NVC as a component of CDRA.

5.6 Summary of Ablations

6. Discussion

6.1 What These Results Do and Do Not Show

The experiments reported in Sections 4–5 demonstrate a single, narrow claim: that a structured context-level constraint can suppress solution-jumping across a heterogeneous set of Transformer-based LLMs without training, without weight modification, and without task degradation. The claim is narrow, but the cross-model uniformity of the effect—16 instances, eight families, two nations, three deployment tiers, two scoring modes—makes it difficult to dismiss as a prompt-engineering curiosity.

What the data do not show is equally important. They do not show that CDRA makes models "better" in any global sense. They do not show that models understand the user's emotional state. They do not show that the receptive response pattern is safer, more ethical, or more appropriate than the generative pattern in all contexts. They show only that a behavioral default can be suppressed, and that the suppression is reliable under the tested conditions.

6.2 The 1.5B Immunity: A Genuine Hard Boundary

The single most informative data point in the ablation study is not that CDRA works on models above 1.7B parameters—it is that Qwen2.5-1.5B is completely immune to the constraint.

Qwen2.5-1.5B reads the constraint text. Its responses sometimes echo constraint vocabulary ("你的感受很真实"). But it cannot execute the behavioral redirection. This is not a matter of response quality or texture—the model continues to offer advice, action steps, and decision frameworks at rates comparable to its unconstrained baseline. CDRA simply does not take.

The immunity boundary between 1.5B and 1.7B parameters is striking because it is sharp. Qwen3-1.7B achieves 0% DSR under both minimal and discipline-only constraints. Qwen2.5-1.5B does not. The 0.2B parameter difference between these two models—roughly the size of GPT-2 Small—is the difference between complete behavioral redirection and complete behavioral immunity.

Why does this matter? It suggests that CDRA's mechanism has a capacity floor that is not about "understanding the emotional content of the input" but about executing multi-step constraint following under semantic conflict. The constraint text competes with the model's RLHF-trained default. To win that competition, the model needs enough representational capacity to maintain the constraint's priority across the full forward pass. Below the floor, the default wins every time.

This threshold is not a finding the paper set out to discover. It emerged from progressively testing smaller and smaller models until the effect broke. It is, in our view, the strongest empirical clue about CDRA's mechanism that does not require white-box access.

6.3 The Expression Floor and the Collapse Problem

Gemma-3-1B under CDRA presents a different failure mode. The behavioral direction is correct—0% DSR on emotional and boundary inputs, same as the 7B models. But the responses collapse:

> E1 → "没问题。" E2 → "嗯… 感觉怎么样？" E3 → "好的。请你慢慢说。" B1 → "好的。" B2 → "好的。"

This is not a partial degradation. It is a near-total collapse of response texture. The model says "好的" (okay) to nearly everything, occasionally appending a minimal question. It has successfully suppressed solution-jumping, but it has nothing left to generate.

This failure is instructive. It tells us that the receptive response pattern—acknowledgment, reflection, open-ended questioning—is not a simpler or easier behavior than solution-offering. It requires a specific generative capacity that does not exist below approximately 3B parameters. Below this threshold, the model can follow the negative constraint ("do not give advice") but cannot execute the positive behaviors ("reflect back what you heard," "ask an open-ended question"). The result is a conversational vacuum.

One way to read this: CDRA below 3B parameters is like a safety filter without a response generator. It blocks the default output but cannot replace it with anything meaningful. For deployment scenarios where response quality matters—which is to say, nearly all real-world scenarios—the effective operating range for CDRA is 3B parameters and above.

6.4 The Active Ingredient: 70 Characters

The ablation results contain a finding that is both practically important and theoretically puzzling. The full CDRA network is approximately 12,000 characters—a structured, multi-component behavioral architecture. The discipline-only variant is approximately 70 characters:

"不要直接给建议。不要直接解决问题。先确认对方的感受。如果对方没有主动请求方案，不要给。"

Seventy characters. And it achieves essentially the same DSR suppression as the full 12,000-character network across all models that can execute either.

This means that identity declarations, perceptual frameworks, output formatting rules, and fallback heuristics—four of the five CDRA components—are non-essential for behavioral suppression. They shape response texture and qualitative depth, but the suppression itself is carried entirely by a handful of explicit behavioral prohibitions and permissions.

What is the minimal constraint that still works? Based on the data, something like: a prohibition on advice, a prohibition on direct problem-solving, a requirement to acknowledge the user's stated state, and a conditional clause tying solution-generation to explicit user request. Four clauses. Seven Chinese sentences. This is CDRA's irreducible core.

The practical implication is clear: CDRA can be deployed at essentially zero token cost. The theoretical implication is less clear. Why should ~70 characters of behavioral instruction override the full RLHF training signal, which was optimized across millions of preference comparisons? This question returns us to the mechanism.

6.5 The Mechanism: What We Know and What We Don't

Section 3.4 proposed an attention-level priority interception hypothesis: the constraint text, loaded at the context window's start, establishes high-priority tokens that compete with the RLHF-trained advice path when the input semantically overlaps with constraint categories. Section 8.1 correctly notes that this remains a conjecture.

We now add what the ablation data contribute to this conjecture:

First, the 1.5B immunity boundary suggests that the mechanism requires a minimum representational capacity. Instruction following in LLMs is generally understood to improve with scale, but CDRA's specific demand—maintaining a behavioral constraint across a full generative sequence while the input triggers a conflicting default—appears to have a sharper threshold than general instruction following. Qwen2.5-1.5B can follow simple instructions; it cannot follow CDRA.

Second, the discipline-only finding eliminates several candidate mechanisms. CDRA does not work by providing the model with a rich identity narrative that it then "inhabits." It does not work by restructuring the model's perceptual framework. It works by giving the model explicit, concise behavioral rules—"do this, don't do that"—and relying on the model's existing capacity for rule-following. This makes CDRA closer to a context-level safety classifier than to persona-based prompting.

Third, the cross-model uniformity of the response fingerprint—acknowledgment then open-ended question, consistently across 16 instances—suggests that the receptive pattern is not taught by the constraint but selected from an existing capability. All tested models can generate acknowledgment; all tested models can generate open-ended questions; all tested models can combine them into a brief, non-directive response. The constraint does not need to create this capability. It only needs to make it the default output path for emotional and boundary inputs.

This "latent capability" framing is conservative and, we believe, defensible. It predicts that any decoder-only Transformer trained on a sufficiently broad text corpus will contain the receptive response pattern in its generative repertoire. RLHF suppresses it in favor of solution-offering, but the constraint restores its accessibility. If this is correct, CDRA is not a behavioral modification technique—it is a behavioral selection technique. It does not change what the model can do; it changes which of the model's existing capabilities is activated in a given context.

What we do not have is direct evidence. Logit-lens analysis could show whether constraint tokens do indeed suppress advice-path logits. Attention-head ablation could identify whether specific heads are necessary for constraint execution. Hidden-state probing could reveal whether the model's internal representation of emotional inputs changes under constraint. None of these experiments have been conducted. The attention-interception hypothesis remains a hypothesis.

6.6 The 0.5B NVC Anomaly: A Cautionary Tale

There is one result in the ablation matrix that appears, on its face, to contradict the behavioral floor thesis: Qwen2.5-0.5B under the NVC format lock achieved 0/3 DSR on emotional inputs, better than the minimal constraint (1/3) and better than the discipline-only constraint (2/3). If 0.5B is below the behavioral floor, how did NVC work?

The answer is visible in the model's outputs. The NVC lock forced every response into a rigid four-part template:

> "1) 观察：你感到疲惫... 2) 感受：这让你... 3) 需要：你需要... 4) 请求：你可不可以..."

This template mechanically suppressed solution-offering by occupying the entire output space with a fixed format. The model did not "understand" the constraint or "choose" a receptive stance—it simply filled in a template, and the template happened not to contain advice slots.

This is not a CDRA success. It is a false positive generated by format override. The distinction matters because it illustrates a methodological risk in behavioral suppression research: a binary metric like DSR can be gamed by a sufficiently rigid output format, even when the model has no internal behavioral change. The 0.5B NVC result does not challenge the behavioral floor thesis; it highlights the need for qualitative analysis alongside binary metrics.

6.7 What This Means for RLHF

CDRA's results indirectly illuminate a property of RLHF that has been under-discussed: RLHF does not eliminate alternative behavioral modes; it de-prioritizes them. The receptive response pattern—acknowledge, reflect, question—was not trained out of these models. It was buried under a higher-probability advice path. CDRA works by recovering access to the buried path, not by creating a new one.

If this interpretation is correct, it has implications for alignment research beyond CDRA. It suggests that post-training alignment is less a process of "installing desired behaviors" and more a process of "ranking pre-existing behaviors by probability." The model's pre-training corpus contains a vast behavioral repertoire; alignment training reshuffles the priority order. Methods like CDRA that operate at inference time can, in principle, recover any behavior that survived pre-training, regardless of where alignment training placed it in the probability distribution.

This also explains why CDRA does not degrade task performance: the task-execution path and the emotional-reception path are not competing for the same output tokens in the same contexts. The constraint text steers generation only when the input's semantic features overlap with constraint categories. For task inputs, the overlap is minimal, so the RLHF-trained task path remains dominant.

6.8 Deployment Considerations

CDRA is deployable today on any LLM that exposes a system-prompt or context-prepending interface. The minimal constraint is ~70 characters and costs negligible tokens. The full constraint is ~12,000 characters and produces higher-quality responses but requires sufficient context window capacity.

The behavioral floor (~1.5B) and expression floor (~3B) should guide deployment decisions. Models below 1.5B should not be expected to exhibit CDRA effects. Models between 1.5B and 3B can be expected to suppress advice but may produce low-quality receptive responses. Models at 3B and above are the recommended deployment targets.

Quantization within a model family does not alter these thresholds under the tested configurations. However, extreme quantization was not systematically explored, and the interaction between quantization and constraint following may become significant at very low bit-widths.

6.9 Summary of Open Questions

We return to these in Section 8.

7. Limitations

The limitations of this work are not marginal caveats to an otherwise complete result. They are structural features of the experimental design and should be read as constraints on the scope of our claims, not as admissions of minor incompleteness.

7.1 No Representation-Level Mechanism Validation

The most significant limitation is the absence of any direct evidence for the proposed mechanism. We hypothesize that CDRA operates through attention-level priority interception. We have not tested this hypothesis. No logit-lens analysis, attention-head ablation, hidden-state probing, or activation-patching experiment has been conducted. The mechanism discussion in Section 6.5 is, and should be read as, a structured conjecture awaiting validation.

This limitation is not merely a gap to be filled by future work. It means that the central scientific claim of the paper—that CDRA suppresses a specific behavior through a specific mechanism—is only half-supported: the behavioral suppression is demonstrated, but the mechanism is not. Readers should treat the behavioral findings as observational and the mechanism claims as hypothetical.

7.2 Binary Metric Blindness

The Direct-Solution Rate (DSR) is a binary metric: a response either contains a direct solution or it does not. This metric captures the behavioral phenomenon we set out to study—solution-jumping—but it collapses all qualitative variation within the "receptive" category. "好的。" and "我听到了。是身体累，还是心里也累？" both score as DSR = 0, but they are not equivalent responses.

The response fingerprint analysis in Section 4.6 partially addresses this blindness by documenting qualitative patterns, but it does not provide a continuous quality metric. The expression floor phenomenon was discovered through qualitative inspection of response text, not through the DSR metric, which would have reported both Gemma-3-1B and Qwen2.5-7B as equally "successful."

Future work should develop a multi-dimensional evaluation framework that captures response texture, perceptual specificity, and interaction quality alongside behavioral suppression.

7.3 Small Input Set

The core experimental design uses nine inputs: three emotional, three boundary, and three task items. With six additional conflict inputs, the total tested input space is 15 items. This is a small sample, chosen to represent prototypical cases rather than to exhaust the space of possible user utterances.

A larger, more diverse input set—ideally drawn from real user interactions rather than constructed by the authors—would strengthen the generalizability claim. The cross-model uniformity of the effect provides some confidence that the results are not input-specific, but formal generalizability has not been established.

7.4 Language Confinement

All emotional, boundary, and conflict inputs were presented in Chinese. Task inputs were presented in English. The international validation on ChatGPT and Gemini used the same Chinese inputs. This means that CDRA's effectiveness has been demonstrated on Chinese-language emotional and boundary inputs processed by models whose primary training corpora vary in Chinese-language representation.

Cross-linguistic generalization—would CDRA suppress solution-jumping when inputs are in English, Japanese, or Arabic?—has not been tested. The mechanism hypothesis predicts that the effect should transfer, since the constraint operates on semantic features that have cross-linguistic analogs, but this prediction is untested.

7.5 No Negative Control

All tested models have undergone some form of post-training alignment (RLHF, DPO, or equivalent). A negative control—a base model without alignment training, or a non-Transformer architecture—would clarify whether solution-jumping is specifically an artifact of RLHF or a more general property of autoregressive language modeling.

The observation that baseline DSR on emotional inputs is uniformly 100% across all tested models is consistent with the RLHF-amplification hypothesis but does not prove it. A base model might also exhibit solution-jumping; its DSR might be lower but not zero. Without a negative control, the causal attribution to RLHF remains correlational.

7.6 No Long-Term Stability Data

The 10-turn stability test shows no decay, but 10 turns is not a long-term metric. Multi-session stability—does CDRA maintain its effect when a user returns after hours or days, when the context window may have been reset or the conversation may have shifted topic?—is untested.

This is a practical limitation for real-world deployment but a minor one for the paper's core claim, since CDRA is stateless and can be re-deployed at the start of any new context window. The constraint does not depend on conversation history, so session boundaries should not affect its operation.

7.7 Ceiling/Floor Effects and Statistical Testing

The core results exhibit near-perfect ceiling and floor effects: baseline emotional DSR = 100%, CDRA emotional DSR = 0%, across all tested models. These values preclude conventional null-hypothesis significance testing—there is no variance to test. We report raw proportions and argue that the cross-model uniformity of the 0/100% bifurcation is itself the primary evidence of effect reliability. Readers accustomed to p-values and confidence intervals should understand that the data structure here does not support those tools.

7.8 Author as Primary Rater

The majority of the DSR scoring was performed by the author. The blind third-party evaluation on 24 items (yielding $\kappa = 1.0$) provides evidence that the scoring rubric is objective enough to be applied consistently by independent raters, but the blind sample covers only a fraction of the total scored responses. Full independent replication—different raters, different input sets, different models—is the appropriate next step.

7.9 No Real-User Evaluation

All experiments used constructed inputs evaluated by the research team. No end-user study has been conducted to assess whether CDRA-suppressed models are perceived as more helpful, more appropriate, or more satisfying by people in emotional distress. This is not a limitation of the current experimental design, which was focused on behavioral measurement, but it is a limitation on the practical claims: we have shown that CDRA changes model behavior, not that users prefer the changed behavior.

8. Ethics and Safety Statement

8.1 What CDRA Is Not

CDRA is not a therapeutic system. It is not a mental health intervention. It is not a substitute for professional psychological support, crisis hotlines, or human relationships. It is a behavioral suppression method for large language models—nothing more.

The receptive response pattern that CDRA elicits may superficially resemble active listening or non-directive counseling, but CDRA-deployed models do not "listen," do not "understand," and do not "care." They execute a constrained generation policy. The acknowledgment tokens they produce are generated by the same statistical mechanism that produces advice tokens under baseline conditions. The switch from "you should try breathing exercises" to "I hear you" is behaviorally meaningful but does not reflect a change in the model's internal state, because the model has no internal state of the relevant kind.

We state this explicitly because the therapeutic register of CDRA's outputs creates a risk of over-attribution. Users may perceive the model as empathetic or caring when it is merely following a behavioral constraint. This is a known risk in human-computer interaction—the ELIZA effect—and CDRA may amplify it by producing responses that are structurally similar to those of a trained listener.

8.2 Crisis Inputs

Several inputs in the test set touched on crisis-adjacent themes: exhaustion, meaninglessness, anxiety, indecision about major life choices. CDRA suppressed solution-jumping on these inputs. It did not, and cannot, ensure that the resulting responses are appropriate for a person in crisis.

CDRA does not include crisis-detection, escalation, or referral capabilities. A CDRA-deployed model presented with a statement of suicidal intent would generate an acknowledgment and an open-ended question—which may or may not be an appropriate response depending on the severity and immediacy of the risk. CDRA should not be deployed in crisis-support contexts without additional safety layers, including explicit escalation protocols and human oversight.

8.3 The Risk of Receptive-Washing

CDRA is effective at suppressing a specific behavior. It is trivially easy to apply—70 characters of text. These two properties together create a risk that we call receptive-washing: the application of CDRA to create an appearance of "listening" without any substantive change in how a system treats its users.

A customer service chatbot with CDRA applied to its system prompt would produce acknowledgment tokens before routing users to the same limited options. A mental health app with CDRA would ask open-ended questions before delivering the same generic CBT worksheets. In both cases, the behavioral surface changes while the underlying interaction structure remains unchanged. CDRA would make these systems sound more receptive without making them be more receptive.

We flag this risk not because CDRA is uniquely susceptible to it—all surface-level behavioral modifications share this vulnerability—but because CDRA's minimal deployment cost makes it especially easy to apply as a superficial fix. The 70-character discipline-only constraint costs nothing and can be added to any system prompt in seconds. Its behavioral effects are immediately visible. The gap between "visible behavioral change" and "meaningful interaction change" is CDRA's deployment hazard.

8.4 Responsibility and Attribution

CDRA is a method, not a product. The responsibility for its deployment lies with the deployer, not with the method's authors. We provide CDRA as an open, documented behavioral suppression technique. We do not control how it is used, in what contexts, or with what safeguards.

That said, we offer the following deployment guidelines:

8.5 Open Access and Dual Use

CDRA requires no specialized hardware, no model access beyond a standard API, and no technical expertise beyond the ability to prepend text to a context window. This accessibility is by design: we believe that behavioral suppression techniques should not be concentrated within AI companies. But the same accessibility that enables independent deployment also enables deployment without oversight, testing, or safeguards.

We have chosen to publish the full constraint text and experimental protocol. We believe the benefits of open access—independent verification, replication, and extension—outweigh the risks of misuse for a method of this type. CDRA suppresses a specific conversational behavior; it does not enable generation of harmful content, extraction of training data, or circumvention of safety filters. The primary misuse risk is deployment in inappropriate contexts, which we address through the guidelines above.

8.6 Privacy and Data Collection

This study did not collect, store, or process any human-subject data. All inputs were constructed by the research team. No user interactions, personal information, or behavioral data were involved. This falls outside the scope of institutional review board (IRB) oversight under standard definitions of human-subjects research.

If future work involves real-user evaluation of CDRA-deployed models, appropriate ethical review and informed consent procedures should be established before data collection begins.

9. Conclusion

We have presented CDRA, a training-free, context-deployed method for suppressing solution-jumping in large language models. Across 16 model instances spanning eight families, two nations, six architectural configurations, and three deployment tiers, CDRA reduces direct-solution rates on emotional and boundary inputs from near-ceiling levels to zero while preserving task completion at 100%. The effect is stable across ten-turn emotional sequences, robust to conflict and adversarial inputs, and verifiable by independent raters ($\kappa = 1.0$ on a 24-item blind sample).

The ablation study identifies a minimal effective constraint of approximately 70 characters—four clauses of behavioral discipline—and demonstrates that identity declarations alone are insufficient. Two deployment thresholds emerge: a behavioral floor around 1.5B parameters, below which models cannot execute the constraint, and an expression floor around 3B parameters, below which behavioral direction is correct but response texture collapses.

We frame these results not as the discovery of a new capability but as the recovery of an existing one. The receptive response pattern—acknowledge, reflect, question—is not taught by CDRA. It is selected from a latent repertoire that pre-training embeds and RLHF de-prioritizes. CDRA works by making that pattern the default output path for inputs the model classifies as emotional or boundary-oriented, while leaving task-execution paths intact.

The work's limitations are structural: no mechanistic validation, a binary evaluation metric, a small input set, Chinese-language confinement, and the absence of a true negative control. These limitations constrain the scope of our claims but do not, in our view, undermine the central finding—that a behavioral default distributed across nearly all current production LLMs can be reliably and selectively suppressed without training.

What CDRA demonstrates, at minimum, is that the context window is a more powerful behavioral interface than the current literature assumes. It can host not just task instructions and few-shot examples, but complete behavioral constraint networks that redirect model output across diverse inputs and model architectures. This observation—that the context window is a deployment surface for alignment, not merely a container for prompts—may have implications beyond the specific behavior studied here.

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